

2	(a) (i) number of moles/amount of substance	B1	[1]
	(ii) kelvin temperature/absolute temperature/thermodynamic temperature	B1	[1]
	(b) $pV = nRT$		
	$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = n \times 8.31 \times 373$	B1	
	$n = 0.38$ (mol)	C1	
	number of molecules or $N = 0.38 \times 6.02 \times 10^{23} = 2.3 \times 10^{23}$	A1	[3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2016	9702	42

or

$$pV = NkT \quad (C1)$$

$$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 373 \quad (M1)$$

$$\text{number of molecules or } N = 2.3 \times 10^{23} \quad (A1)$$

$$(c) \text{ volume occupied by one molecule} = (2.4 \times 10^3) / (2.3 \times 10^{23}) \quad C1$$

$$= 1.04 \times 10^{-20} \text{ cm}^3$$

$$\text{mean spacing} = (1.04 \times 10^{-20})^{1/3} \quad C1$$

$$= 2.2 \times 10^{-7} \text{ cm (allow 1 s.f.)} \quad A1 \quad [3]$$

(allow other dimensionally correct methods e.g. $V = (4/3)\pi r^3$)